

Indefinite Integrals

Evaluate each indefinite integral.

$$1) \int 30x^5 dx$$
$$5x^6 + C$$

$$2) \int 24x^5 dx$$
$$4x^6 + C$$

$$3) \int -3 dx$$
$$-3x + C$$

$$4) \int 15x^2 dx$$
$$5x^3 + C$$

$$5) \int (12x^5 - 6x) dx$$
$$2x^6 - 3x^2 + C$$

$$6) \int (20x^3 + 4x) dx$$
$$5x^4 + 2x^2 + C$$

$$7) \int (12x^5 - 4x) dx$$
$$2x^6 - 2x^2 + C$$

$$8) \int (24x^5 + 1) dx$$
$$4x^6 + x + C$$

Critical thinking question:

9) What is the derivative of $2x^6 - 2x^2 + C$?

$$12x^5 - 4x$$